

BAYAN, Mongolia

WMO: 442940

Lat: 47.2567N

Lon: 107.5378E

Elev: 1430

StdP: 85.28

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.0	-31.9	-37.4	0.1	-33.5	-35.2	0.2	-31.4	12.0	-20.3	10.4	-20.2	1.5	0	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	29.6	16.0	27.4	15.1	25.5	14.4	17.7	26.1	16.7	24.5	15.8	23.0	4.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	12.7	20.2	13.9	11.8	19.7	12.9	11.0	19.0	56.1	25.8	52.5	24.6	49.5	23.1	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	10.4	9.6	-37.2	32.7	2.6	2.4	-39.1	34.4	-40.7	35.8	-42.2	37.2	-44.1	38.9
			-37.1	19.8	2.6	1.9	-39.0	21.1	-40.6	22.2	-42.0	23.3	-43.9	24.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-22.2	-18.1	-8.4	1.6	9.1	15.6	18.4	15.9	9.1	-0.4	-12.7	-20.0	(6)
(7)		DBStd	15.11	4.37	5.41	6.27	5.19	4.88	4.05	3.38	3.75	4.66	4.95	6.78	4.85	(7)
(8)		HDD10.0	4705	998	788	572	254	77	5	0	3	72	324	682	932	(8)
(9)		HDD18.3	7114	1256	1021	830	501	289	100	40	94	279	582	932	1190	(9)
(10)		CDD10.0	713	0	0	0	3	48	173	261	186	43	0	0	0	(10)
(11)		CDD18.3	81	0	0	0	0	2	18	42	18	1	0	0	0	(11)
(12)		CDH23.3	1000	0	0	0	1	55	266	447	215	17	0	0	0	(12)
(13)		CDH26.7	271	0	0	0	0	9	73	134	53	1	0	0	0	(13)
(14)	Wind	WSAvg	3.6	2.7	2.9	3.8	4.8	5.2	4.3	3.7	3.4	3.6	3.4	2.8	2.5	(14)
(15)	Precipitation	PrecAvg	223	1	1	2	6	15	42	66	58	24	6	4	1	(15)
(16)		PrecMax	369	5	7	10	25	86	184	132	197	73	19	14	5	(16)
(17)		PrecMin	132	0	0	0	0	0	0	17	7	0	0	0	0	(17)
(18)		PrecStd	59	1	1	2	6	16	32	31	36	17	6	4	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.1	0.3	11.6	20.9	28.0	31.6	33.0	31.0	25.5	17.2	6.6	-5.4	(19)
(20)		MCWB	-8.1	-3.5	3.1	8.9	12.1	15.7	18.1	17.2	13.0	7.1	1.3	-7.3	(20)	
(21)		2%	DB	-9.7	-2.7	7.8	17.0	24.8	28.8	30.2	27.8	22.9	13.9	3.4	-8.2	(21)
(22)		MCWB	-11.2	-5.9	1.2	6.6	11.9	14.5	17.0	15.6	12.3	6.1	-0.8	-9.6	(22)	
(23)		5%	DB	-12.2	-5.5	5.0	14.3	22.2	26.3	27.7	25.6	20.6	11.3	0.6	-10.3	(23)
(24)		MCWB	-13.3	-7.6	-0.4	5.5	10.3	13.6	16.1	14.9	11.2	4.4	-2.6	-11.4	(24)	
(25)		10%	DB	-14.6	-8.2	2.1	11.7	19.6	24.1	25.7	23.5	18.3	8.9	-2.0	-12.2	(25)
(26)		MCWB	-15.4	-9.8	-2.2	4.1	9.2	13.1	15.4	14.2	10.1	3.0	-4.4	-13.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.0	-3.2	3.5	10.2	15.1	17.7	20.2	18.5	14.9	8.3	1.5	-7.0	(27)
(28)		MCDWB	-6.2	0.0	10.5	19.4	25.1	26.7	29.6	26.5	21.8	15.8	6.5	-5.5	(28)	
(29)		2%	WB	-11.1	-5.8	1.4	7.4	12.7	16.2	18.5	17.1	13.1	6.3	-0.6	-9.5	(29)
(30)		MCDWB	-9.7	-3.0	7.3	15.4	22.5	25.5	27.1	25.3	21.0	12.8	3.0	-8.3	(30)	
(31)		5%	WB	-13.3	-7.6	-0.4	5.7	11.0	15.1	17.2	16.0	11.9	4.7	-2.4	-11.3	(31)
(32)		MCDWB	-12.3	-5.5	4.6	13.6	20.3	23.7	25.1	23.1	19.1	11.0	0.5	-10.2	(32)	
(33)		10%	WB	-15.4	-9.8	-2.2	4.2	9.6	14.0	16.3	15.0	10.6	3.2	-4.4	-13.1	(33)
(34)		MCDWB	-14.6	-8.2	1.9	11.6	18.3	21.6	23.5	21.9	17.5	8.5	-2.0	-12.2	(34)	
(35)	Mean Daily Temperature Range		MDBR	12.9	14.9	14.7	14.4	14.5	12.7	11.6	12.3	14.2	14.1	13.2	12.0	(35)
(36)		5% DB	MCDBR	15.7	17.9	17.9	18.5	19.1	16.1	14.8	15.6	17.7	18.5	16.1	13.3	(36)
(37)		MCWBR	14.3	15.3	12.3	11.1	10.1	7.1	6.1	6.8	9.0	11.8	12.4	12.0	(37)	
(38)		5% WB	MCDBR	15.5	17.6	17.7	17.6	17.1	14.1	12.8	13.3	15.5	17.5	15.7	13.1	(38)
(39)		MCWBR	14.2	15.1	12.4	11.3	10.2	7.2	6.3	6.7	8.5	11.4	12.3	11.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.236	0.260	0.298	0.345	0.379	0.392	0.372	0.352	0.310	0.280	0.251	0.237	(40)	
(41)		taud	2.368	2.325	2.286	2.223	2.166	2.228	2.344	2.376	2.432	2.457	2.413	2.362	(41)	
(42)		Ebn at Noon	877	919	925	906	883	871	885	890	904	886	848	829	(42)	
(43)		Edh at Noon	68	89	110	130	143	136	119	111	95	78	65	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.98	3.12	4.62	5.88	6.81	6.72	6.33	5.69	4.79	3.35	2.12	1.57	(44)	
(45)		RadStd	0.14	0.21	0.17	0.24	0.41	0.36	0.34	0.33	0.23	0.17	0.13	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	-2.13	N/A	-0.68	-0.65	N/A	N/A	N/A	N/A

Nomenclature: See separate page